Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C008 **TOR** 4 genes / 4 species 6 direct pairs	**Y1 TOR1** GenAge 679 **Reviewed example** Deletion **Replicative life ↑** PMID 16293764	**W1 let-363** GenAge 32 **Reviewed example** RNAi **Adult life ↑** PMID 14668850	**F1 Tor** GenAge 678 **Reviewed example** Dominant-negative **Adult life ↑** PMID 15186745	**M1 Mtor** GenAge 1809 **Reviewed example** Hypomorph **Median survival ↑** PMID 23994476

Direct pairs: W1–F1 (1:1); W1–M1 (1:1); W1–Y1 (1:m); F1–M1 (1:1); F1–Y1 (1:m); M1–Y1 (1:m)

Yeast replicative lifespan differs from animal survival. Yeast TOR1 pairs are non-1:1; TOR and S6K share a pathway.

C073 **S6 kinase** 3 genes / 3 species 2 direct pairs	No gene in this candidate network	**W1 rsks-1** GenAge 578 **Reviewed example** RNAi **Mean life ↑** PMID 17266680	**F1 S6k** GenAge 583 **Reviewed example** Dominant-negative **Adult life ↑** PMID 15186745	**M1 Rps6kb1** GenAge 1737 **Reviewed example** Global knockout **Female life ↑*** PMID 19797661

Direct pairs: W1–F1 (1:1); F1–M1 (1:1)

* Mouse benefit is female-specific; male increase was not significant. The 2011 correction concerns a worm strain, not mouse survival. No worm–mouse pair is implied.

C019 **Insulin/IGF receptor** 4 genes / 3 species 3 direct pairs	No gene in this candidate network	**W1 daf-2** GenAge 163 **Reviewed example** Hypomorph **Adult life ↑** PMID 8247153	**F1 InR** GenAge 368 **Reviewed example** Hypomorph **Female life ↑** PMID 11292875	**M1 Igf1r** GenAge 365 **Reviewed example** Heterozygous KO **Female life ↑*†** PMID 12483226 **M2 Insr** GenAge 373 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12543978, 15713666

Direct pairs: W1–F1 (1:1); W1–M1 (1:n); W1–M2 (1:n)

* Original female Igf1r benefit; male increase not significant. † Not reproduced in a later cohort (PMID 22132081).

Insr is a separate paralog; no fly–mouse pair is implied.

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.

Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.

GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.

Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.